

SOC176 - RDP Brute Force Detected

Cybersecurity Portfolio Project

Project Summary

Field	Value
Project Type	SIEM Alert Investigation / Brute Force Login Investigation
Platform	LetsDefend SOC practice lab
Alert Rule	SOC176 - RDP Brute Force Detected
Primary Target	Host Matthew - 172.16.17[.]148
Attacker Source IP	218.92.0[.]56
Protocol / Service	RDP - TCP/3389
Assessment	True Positive - successful brute-force login followed by post-authentication activity
Severity	High for portfolio report; original lab pages may display Low/Medium depending on alert instance

Title:

SOC176 - RDP Brute Force Detected: SIEM Alert Investigation

Scenario:

A LetsDefend SOC alert identified suspicious RDP authentication activity against the host Matthew (172.16.17[.]148). The activity originated from an external IP address, 218.92.0[.]56, and involved repeated failed login attempts using common usernames. The investigation objective was to validate whether the alert represented a false positive, an unsuccessful attack, or a successful brute-force compromise.

Tools used:

- LetsDefend SIEM / Monitoring Dashboard
- LetsDefend Log Management
- LetsDefend Endpoint Security
- LetsDefend Threat Intelligence
- VirusTotal / AbuseIPDB / public IP reputation checks
- Windows Security Event Logs - Event ID 4625 and 4624
- MITRE ATT&CK Framework

Alert / initial evidence:

Alert Description: SOC176 - RDP Brute Force Detected.

The alert was triggered by multiple failed RDP authentication attempts from a single external source. Public walkthrough evidence for the Event ID 234 version of the lab documents the source IP as 218.92.0[.]56, target host Matthew, destination IP 172.16.17[.]148, and protocol RDP. The trigger reason was login failures from one source using different/nonexistent accounts.

Investigation steps:

- Reviewed the SOC176 alert overview in LetsDefend Monitoring.
- Took ownership of the alert and created/opened the investigation case.
- Confirmed that the source IP address was external, not an internal corporate address.
- Performed IP reputation checks using LetsDefend Threat Intel and external sources such as VirusTotal / AbuseIPDB.
- Filtered Log Management using the source IP 218.92.0[.]56 and destination host/IP Matthew - 172.16.17[.]148.
- Reviewed Windows authentication logs for failed logins (Event ID 4625).
- Checked for successful authentication after the failed attempts (Event ID 4624).
- Confirmed RDP service targeting on TCP/3389.
- Reviewed Endpoint Security for post-authentication activity and terminal/CMD history.
- Documented IOCs, mapped observed behaviour to MITRE ATT&CK, and selected containment/remediation actions.

Findings:

Finding Area	Evidence / Analyst Notes
Failed login attempts analysis	Public walkthroughs document repeated failed RDP logins from 218.92.0[.]56 against Matthew. One walkthrough reports 14 failed attempts followed by one successful login; another reports 29 failed Event ID 4625 events and one successful login event. Your own lab screenshots should use the count shown in your LetsDefend instance.
Timeline of event	The public Event ID 234 walkthrough notes a successful login to Matthew at 08:44 AM after failed attempts. Post-authentication activity in another walkthrough includes commands around 11:45-11:46 such as cmd.exe, whoami, net user, net localgroup administrators, and netstat -ano.
Source IP / username analysis	Source IP 218.92.0[.]56 was external and reportedly malicious. Usernames attempted included common/default-style accounts such as admin, guest, sysadmin, and Matthew.
Successful login check	A successful login after the failed attempts was observed, making this a true-positive compromise rather than only attempted brute force.
False positive / true positive assessment	True Positive. Repeated failed logins, RDP targeting, malicious external source IP reputation, successful authentication, and post-authentication discovery activity support malicious activity.
Severity	High in portfolio context because the brute-force attack succeeded and post-authentication discovery activity was observed. The public LetsDefend alert page may list a different severity for another SOC176 instance.

Indicators of Compromise:

- Source IP: 218.92.0[.]56
- Target Host: Matthew
- Target IP: 172.16.17[.]148
- Service: RDP / TCP 3389
- Authentication failures: Windows Event ID 4625
- Successful authentication: Windows Event ID 4624
- Attempted usernames: admin, guest, sysadmin, Matthew
- Post-authentication commands: cmd.exe, whoami, net user, net localgroup administrators, netstat -ano

MITRE ATT&CK mapping:

- T1110 - Brute Force: repeated authentication attempts against RDP.
- T1078 - Valid Accounts: successful login after failed attempts indicates possible use of valid credentials.
- T1021.001 - Remote Services: Remote Desktop Protocol: RDP was the targeted remote access service.
- T1033 - System Owner/User Discovery: whoami command observed in post-authentication activity.
- T1087.001 - Account Discovery: Local Account: net user letsdefend activity observed.
- T1069.001 - Permission Groups Discovery: Local Groups: net localgroup administrators activity observed.
- T1049 - System Network Connections Discovery: netstat -ano activity observed.

Impact assessment:

The attack should be treated as a confirmed compromise because the failed login pattern was followed by successful authentication. Post-authentication commands suggest early discovery activity after access was obtained. Potential impact includes unauthorized access to the Matthew host, credential exposure, privilege escalation attempts, persistence preparation, and lateral movement risk.

Recommended actions:

- Escalate to Tier 2 SOC / Incident Response because successful login was observed.
- Isolate the Matthew endpoint to prevent further activity.
- Block or restrict the malicious source IP 218.92.0[.]56 at perimeter controls where appropriate.
- Reset the affected user credentials and review all accounts attempted during the attack.
- Enable MFA for RDP and all remote-access paths.
- Implement account lockout after repeated failed login attempts.
- Restrict RDP exposure to VPN, internal networks, or approved IP allowlists only.
- Increase SIEM alert thresholds and correlation rules for repeated 4625 events followed by 4624 success.
- Review endpoint for persistence, new users, suspicious scheduled tasks, and unusual processes.
- Document the event, IOCs, analyst notes, and final classification as True Positive.

Conclusion:

The SOC176 alert was validated as a true-positive RDP brute-force incident. The investigation confirmed repeated failed authentication attempts from an external malicious IP, targeting the Matthew host over RDP, followed by successful login and post-authentication discovery activity. The incident demonstrates SIEM triage, log filtering, authentication analysis, threat intelligence enrichment, endpoint review, MITRE mapping, and escalation decision-making - all suitable for a junior SOC analyst portfolio.